

JADE OS · Benefits & Features

Technical Brief

Prepared by JADE OS · Operations Workbench · June 07, 2026

This document is operator-grade. Every number is benchmark-traceable; no synthetic data appears.

1 · Executive Overview

JADE OS is the operator-grade AI layer that composes freight-domain agents into a single, auditable workflow. This brief documents what ships in production today, what's pilot-phase (deterministic MVP live; full ML/ingest layer activates during a paid pilot), and what's on the Phase-4 Scale roadmap.

- Module ship status · 1 LIVE (M4 Pricing/Rate-Floor) · 2 LIVE-PARTIAL (M2 Route, M3 Compliance) · 3 PILOT-PHASE (M1 Dispatch, M5 Retention, M6 Maintenance)
- Cross-cutting substrate · workflow memory, claims, rate floors, immutable audit chain — all LIVE
- 90-day pilot template · 4-phase rollout · success metrics declared pre-pilot
- Honest framing · M1/M5/M6 ship as deterministic MVPs today; full LLM coaching + survival models land in Pilot
- Built on the same workflow-memory + claims/risk substrate you can audit in the console

2 · Module M1 · Dispatch Optimizer · Deep Dive (PILOT-PHASE)

Status today · Deterministic MVP ships at POST /api/agent/dispatch/recommend. Scores every candidate driver-load pairing using constraint-satisfaction + utility weighting (cost, HOS feasibility, customer priority, driver lane preference). Returns top-3 per load.

- Inputs (live MVP) · driver lat/lon + HOS state + lane prefs · load O/D + window + priority + revenue
- Decision (live MVP) · cost·priority·deadhead·HOS-feasibility utility score · deterministic, reproducible
- Action (Pilot) · TMS write-back via Descartes/McLeod/TMW APIs · currently returns recommendations only
- KPIs · empty-mile reduction 8-15% · dispatcher load -30-50% · time-to-dispatch -40% (target band, validated in pilot)
- Pilot-phase add-ons · LLM dispatcher-coaching prompts · real-time ELD position ingest · live load-board feed

3 · Module M2 · Route + Fuel Agent · Deep Dive (LIVE · PARTIAL)

Status today · Live routing at POST /api/trucker/route (OSRM) + truck-stops + fuel/parking at POST /api/trucker/truck-stops (OpenStreetMap Overpass). Weather + DOT 511 directory + state-truck restriction warnings ship today.

- Inputs (live) · O/D, OSM stops, NOAA weather, 50-state DOT 511 directory
- Decision (live) · graph routing via OSRM

- Action (live) · returns route + qualifying stops + weather; explicit warning that auto profile must be cross-checked vs state truck restrictions
- KPIs · fuel savings target 10-18% · validated when MILP + EIA daily diesel land in Pilot
- Pilot-phase add-ons · fuel-arbitrage MILP picking cheapest qualifying station · EIA daily diesel feed (key required) · driver loyalty-stop preference

4 · Module M3 · Compliance + Safety · Deep Dive (LIVE · PARTIAL)

Status today · FMCSA §395 HOS rules are codified locally; POST /api/trucker/hos-check returns violations cited to paragraph + remaining drive/duty/break hours. /api/trucker/hos-rules returns the federal limits dict.

- Inputs (live) · driver HOS state (driving hours so far, on-duty hours so far, last break)
- Decision (live) · rules engine on FMCSA Part 395 · violations cited to §395.3(a)(2)/(3) etc.
- Action (live) · returns issues list + remaining hours; integrates with M1 dispatch feasibility
- KPIs · violation drop 40-60% target · audit-pack prep · days → hours (validated in pilot)
- Pilot-phase add-ons · DOT-audit-ready PDF pack generator · DQF (Driver Qualification File) ingest · §396 vehicle inspection rules engine · CSA-score trend dashboard

5 · Module M4 · Pricing + Rate-Floor Guard · Deep Dive (LIVE · PRODUCTION)

Status today · LIVE in production. POST /api/quotes/validate runs every quote against a layered floor (manual + formula + historical P25), assigns LOW/MEDIUM/HIGH/CRITICAL severity, and emits AUTO_OK / QUEUE_REVIEW / HARD_BLOCK. HARD_BLOCK can only be cleared by an authorized override with written notes — both the block and the override are immutably audit-chained.

- Inputs (live) · proposed sell rate · carrier pay · lane code · equipment · cost basis
- Decision (live) · max(manual, formula, historical_p25) floor; severity bands; decision matrix
- Action (live) · persists quote_review row + audit event regardless of outcome; SLA timer per severity
- KPIs · margin uplift 1.5-3.0 pp · 100% of below-floor quote attempts blocked
- Hard block · only an authorized role can override; override is immutably audited

6 · Module M5 · Driver Lifecycle · Deep Dive (PILOT-PHASE)

Status today · Deterministic MVP ships at POST /api/agent/retention/risk. Scores 0-100 with transparent weighted factors (home-time deficit, pay stagnation, tenure cliffs, complaints, safety events, mileage

outliers). Returns LOW/MEDIUM/HIGH/CRITICAL band + top factors + actionable recommendations.

- Inputs (live MVP) · tenure_months, miles_last_30d, home_time_days_last_30d, weeks_since_last_pay_raise, dispatch_complaints_90d, safety events
- Decision (live MVP) · deterministic weighted scoring with documented coefficients · reproducible
- Action (live MVP) · weekly retention queue with recommended action per driver
- KPIs · 12-mo retention +6-12 pp · turnover cost saved \$400-900 per driver (target, validated in pilot)
- Pilot-phase add-ons · proper survival analysis (Cox proportional hazards) · LLM coaching template grounded in last 90 days of activity · NPS pulse integration

7 · Module M6 · Predictive Maintenance + Carrier Match · Deep Dive (PILOT-PHASE)

Status today · Deterministic MVP ships at POST /api/agent/maintenance/window. Uses a J1939 SPN severity dictionary + PM mileage thresholds (PM-A 10k, PM-B 40k) + inspection-violation count to return urgency 0-100, band (ROUTINE/WATCH/SCHEDULE_SOON/GROUND_NOW), window in days, and specific recommended services.

- Inputs (live MVP) · vehicle_id · miles since PM-A/PM-B · fault codes (last 30d) · violation count
- Decision (live MVP) · severity table + mileage thresholds + long-haul amplifier
- Action (live MVP) · returns service list + recommended window; ready for manual shop scheduling
- KPIs · unplanned downtime -20-30% target
- Pilot-phase add-ons · live J1939 ingest via Geotab/Samsara API · time-series anomaly detection · broker-side carrier scorecard

8 · ROI Analysis · Two Case Studies

Two illustrative archetypes drawn from public mid-market freight benchmarks (ATA · ATRI · BLS · FMCSA · EIA). ROI assumes M4 (Pricing) in production from day 1 + M1/M3 deterministic-MVPs running shadow-mode in month 1, shifting to L2 production by month 3.

Case A · Small Regional · 50 trucks

- Annual savings ~ \$380-460k · upfront \$38k · license \$60k
- Payback 6-8 months · 3-yr NPV ~ \$720k @ 10% discount

Case B · Mid-Market · 175 trucks

- Annual savings ~ \$2.0-2.4M · upfront \$130k · license \$210k
- Payback 2-3 months · 3-yr NPV ~ \$4.5M @ 10% discount

Sources · ATA Trucking Industry Profile · ATRI Operational Costs of Trucking 2024 · BLS OEWS 53-3032
· EIA weekly diesel · FMCSA Safety Progress Report

9 · Implementation Roadmap · Four Phases

- Phase 1 · Readiness · 1 week · data inventory, integrations confirmed, success metrics signed in writing
- Phase 2 · Baseline · 4 weeks · M4 (Pricing) production · M1+M3 shadow mode · baseline KPIs measured
- Phase 3 · Pilot · 8-12 weeks · M1/M3 in L1/L2 production, M5/M6 deterministic MVPs activated · weekly KPI review · LLM coaching layers wired progressively
- Phase 4 · Scale · ongoing · M2 fuel MILP + live ELD/TMS write-back + M6 J1939 ingest · quarterly business review · audit refresh

10 · Minnesota Opportunity & Technical Appendix

Minnesota's freight base — Twin Cities I-94/I-35 corridor, intermodal at Midway/CP terminals, agricultural outbound, and a deep mid-market fleet roster (Bay & Bay, Dart, Halvor Lines, ATS, Lakeville Motor Express, and dozens of 50-250 truck operators) — is a prime mid-market beachhead. Adoption tends to follow proof, so pilots are structured for fast metric capture.

Technical Appendix · Live APIs · POST /api/quotes/validate · POST /api/agent/dispatch/recommend · POST /api/trucker/route · POST /api/trucker/hos-check · POST /api/agent/retention/risk · POST /api/agent/maintenance/window · GET /api/agent/modules/status (honest ship-status manifest) · Auth model (admin JWT today, tenant JWT in multi-tenant rollout) · Audit substrate (immutable SHA-256 hash chain, daily Merkle-anchor option) · LLM substrate (Claude Sonnet 4.5 + GPT-5.2 via Emergent · router fallbacks · rate-limit + budget alerts).

— end of brief · sources cited inline —